04-12 第一篇

Biomimicry: Learning from Nature's Designs

Have you ever wondered how a tiny gecko can stick to a smooth wall or why a kingfisher's beak is shaped the way it is? Scientists and engineers are increasingly looking to nature for inspiration when designing new products and solving problems. [1] This approach is called biomimicry—a word that comes from "bio" (life) and "mimicry" (copying). Instead of inventing things from scratch, designers study how plants, animals, and even bacteria have solved similar challenges over millions of years. Nature's solutions are often more efficient, use less energy, and produce less waste than human-made ones.

One famous example is Velcro. In the 1940s, a Swiss engineer named George de Mestral noticed that burrs—small hooks from plants—stuck tightly to his clothes and his dog's fur when he returned from a walk. [2] He examined them under a microscope and saw that each burr had thousands of tiny hooks that caught on loops in the fabric. He then created a material with two sides: one covered in small hooks and the other in soft loops. When pressed together, they stick just like the burrs on his clothes. Today, Velcro is used everywhere, from shoes and bags to space suits.

Transport is another area where nature has inspired better design. Japan's high-speed "bullet train" used to create a loud boom when it entered tunnels because of the way air was pushed ahead of it. Engineers studied the kingfisher, a bird that dives into water with almost no splash. [3] The bird's long, pointed beak allows it to cut through the water smoothly. By reshaping the front of the train to look like the kingfisher's beak, engineers reduced the noise and made the train faster and more energy-efficient. This one change saved both energy and comfort for people living near the tracks.

Buildings and materials are also being improved through biomimicry. [4] The lotus plant has leaves that stay clean even in muddy water because their surface is covered in tiny bumps that cause water and dirt to roll off. Paint and fabric companies have copied this structure to create "self-cleaning" surfaces that need less washing and fewer chemicals. Similarly, termite mounds in Africa stay cool inside even when the outside temperature is very high. Architects have studied their ventilation system to design buildings that use less air conditioning.

[5] From sticky tape to high-speed trains and self-cleaning windows, nature continues to teach us better ways to live and build. Perhaps the best lesson of biomimicry is that we do not have to choose between progress and the environment—we can design with both in mind.

选项

A. In short, biomimicry shows that the best ideas may already exist in the natural world around us. B. The key is to observe carefully and then adapt what works in nature for human use. C. Its slender shape reduces resistance and prevents the kind of pressure buildup that caused the train's problem. D. For example, lotus leaves and termite mounds have inspired buildings and materials that use less energy. E. He wondered if he could copy this natural "sticking" ability to make something useful for people. F. By studying how living things have solved problems, we can create products that work better and harm the planet less.

答案键（Answer Key）

[1] F [2] E [3] C [4] D [5] A

深度解析

- **[1] F**：
  - **线索**：定义后概括与引出。空前介绍"biomimicry"的词源及"Instead of inventing things from scratch, designers study how plants, animals...have solved similar challenges"。F选项"By studying how living things have solved problems, we can create products that work better and harm the planet less"（通过研究生物如何解决问题，我们能创造出效果更好、对地球伤害更小的产品）中"studying how living things have solved problems"与空后"designers study how...have solved"呼应，"work better and harm the planet less"与空后"more efficient, use less energy, produce less waste"对应，自然承接定义并引向具体案例。
- **[2] E**：
  - **线索**：观察发现与探究动机。空前描述"Mestral noticed that burrs...stuck tightly to his clothes and his dog's fur"，空后"He examined them under a microscope...He then created a material...stick just like the burrs"。E选项"He wondered if he could copy this natural 'sticking' ability to make something useful for people"（他想知道能否复制这种自然的"黏附"能力来为人类制造有用的东西）中"copy this natural sticking ability"与空前"burrs stuck tightly"及空后"created a material...stick just like the burrs"形成因果链，"make something useful"自然引出空后创造 Velcro 的过程，衔接顺畅。
- **[3] C**：
  - **线索**：鸟嘴结构与工程应用。空前介绍"Engineers studied the kingfisher...dives into water with almost no splash"，空后"By reshaping the front of the train to look like the kingfisher's beak, engineers reduced the noise and made the train faster"。C选项"Its slender shape reduces resistance and prevents the kind of pressure buildup that caused the train's problem"（其修长外形减少阻力并防止导致列车问题的那种压力积聚）中"Its slender shape"指代 kingfisher's

beak，"reduces resistance"与"no splash...smoothly"对应，"pressure buildup that caused the train's problem"与空前"loud boom...air was pushed ahead"及空后"reduced the noise"完美契合，逻辑紧密。

- [4] D：
  - **线索**：建筑材料与自然启发总领。空前提出"Buildings and materials are also being improved through biomimicry"，空后分别详述"lotus plant...self-cleaning surfaces"与"termite mounds...ventilation system...less air conditioning"。D选项"For example, lotus leaves and termite mounds have inspired buildings and materials that use less energy"（例如，荷叶和白蚁丘启发了耗能更少的建筑与材料）中"For example"引出例证，"lotus leaves and termite mounds"与空后两段具体描述一一对应，"use less energy"概括二者共同效果，衔接自然。
- [5] A：
  - **线索**：全文收束与升华。空[5]位于结尾段段首，空后列举"sticky tape, high-speed trains, self-cleaning windows"并总结"nature continues to teach us...we can design with both in mind"。A选项"In short, biomimicry shows that the best ideas may already exist in the natural world around us"（简言之，仿生学表明最好的创意可能早已存在于我们周围的自然世界中）用"In short"总结全文，"best ideas...in the natural world"与空后"nature continues to teach us"呼应，"around us"与"design with both in mind"形成升华，收束得当。
- **干扰项分析**：
  - **B. The key is to observe carefully and then adapt what works in nature for human use.**：强调观察与 adaptation，与仿生学主题相关，易被误填[1]或[2]。但空[1]需概括"by studying...we can create"的因果关系，B侧重方法论（observe and adapt），与空后"more efficient, use less energy"等结果性描述衔接不及F直接；空[2]空前为具体观察（burrs stuck），空后为微观检验与创造，E的"wondered if he could copy"直接引出探究动机，B的"observe and adapt"过于笼统。属"主题相关但功能错位"型干扰。